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CAREER SUMMARY

I specialise in characterisation of materials using X-ray methods, particularly X-ray absorption spectroscopy and small-angle X-ray scattering. In graduate school, I studied light-induced processes in zirconium-based metal–organic frameworks, focusing on charge transfer processes. In my postdoc, I refined expertise in X-ray methods by studying structural properties of disordered actinide oxides and salts. I am targeting an eventual transition into a staff scientist role. I have full working rights in Australia.

EDUCATION

Ph.D. , Inorganic & Physical Chemistry	Northwestern University, Evanston, IL, USA	2018 – 2023
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Dissertation title: The Relationship between Node Ligand Environment and Excited State Dynamics of Zirconium-Based Metal–Organic Frameworks.

Advisors: Prof. Joseph T. Hupp, Prof. Lin X. Chen

B.S. , Chemistry	Rensselaer Polytechnic Institute, Troy, NY, USA	2014 – 2018
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RESEARCH EXPERIENCE

Postdoctoral Scholar at Moscow State University, Moscow, Russia	Mar. 2024 – Dec. 2025
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Summary: Research activity within the Radiochemistry Division, Department of Chemistry, advised by Dr Anna Romanchuk and Professor Stepan Kalmykov. Conducted aqueous chemistry under controlled pH & ionic strength conditions, characterised diverse inorganic products with X-ray methods. 2 publications, 1 in preparation.

Research Theme: Speciation of hazardous radionuclides in the natural environment.

Key Achievements:

- Characterised morphology in a series of amorphous thorium oxides by applying a combination of small-angle X-ray scattering, X-ray diffraction, X-ray pair distribution function, transmission electron microscopy and mass fractal aggregate growth simulations
- Described speciation of thorium in an environment mimicking agricultural runoff: isolated hydrated sodium–thorium phosphates, characterised their structure and chemical composition by applying electron microscopy and X-ray absorption spectroscopy (DOI: [10.1039/D5DT00276A](https://doi.org/10.1039/D5DT00276A))
- Applied extended X-ray absorption fine structure spectroscopy, including Reverse Monte-Carlo analysis, to confirm local structure around neptunium and plutonium cations in phosphates and carbonates forming under conditions mimicking a nuclear waste storage breakthrough scenario (DOI: [10.1039/D5DT00913H](https://doi.org/10.1039/D5DT00913H)).
- Developed a synthetic protocol to synthesise known hydrated cerium phosphates (proxies for plutonium phosphates) with deuterons instead of protons to facilitate a neutron diffraction structural study
- Prepared and completed 3 successful experimental proposals for SAXS and neutron diffraction experiments at large-scale user facilities within the Kurchatov Institute National Research Centre (Moscow, Russia) and the Joint Institute for Nuclear Research (Dubna, Russia)

Graduate Research Assistant at Northwestern University, Evanston, IL, USA	Sep. 2018 – Dec. 2023
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Summary: Research activity within the Department of Chemistry, jointly supervised by Professors Joseph Hupp and Lin Chen. Carried out air- and water-sensitive organic and inorganic synthesis. Characterised metal–organic frameworks with X-ray and optical spectroscopy. Managed instrumentation. 8 publications, 1 submitted.

Research Theme: Development of photoactive zirconium-based metal–organic frameworks for renewable energy.

Key Achievements:

- Modelled excited state evolution as a function of acidity of the aqueous environment for a zirconium-based metal–organic framework by applying target kinetic analysis; based on the results, suggested a hole transfer based mechanism for the formation of the excited state (DOI: [10.1021/acs.jpcclett.4c02848](https://doi.org/10.1021/acs.jpcclett.4c02848))

- Modelled excited state evolution in zirconium-based metal—organic frameworks carrying single atom catalyst mimics: studied 1st and 2nd order dynamic processes in the solid state (excitation energy transfer, charge transfer), assessed the structure-activity relationship (DOI: [10.1021/acs.energyfuels.1c02623](https://doi.org/10.1021/acs.energyfuels.1c02623))
- Designed a specialised experimental electrolytic cell to use for *operando* X-ray absorption experiments at a synchrotron beamline; characterised electronic structure of nanoscale core-shell titanium dioxide catalysts (DOI: [10.1021/acs.chemmater.4c02190](https://doi.org/10.1021/acs.chemmater.4c02190))
- Drafted and published an Account (self-review) describing results obtained by Professor Lin X. Chen's research group in the field of excitonic properties of framework solids (DOI: [10.1021/accounts.2c00137](https://doi.org/10.1021/accounts.2c00137))
- Managed and ensured safe operation of scientific instrumentation (Ar-filled glovebox (VAC), supercritical CO₂ manifold (Tousimis), UV-Vis with an integrating sphere (Shimadzu), fluorimeter (HORIBA Fluorolog); trained 20+ colleagues and drafted standard operating procedures

SKILLS AND EXPERTISE

- **Laboratory:** hydrothermal synthesis, air-free methods, Schlenk line operation, chemical waste management
- **Characterisation of materials:** XANES, EXAFS, SAXS, powder XRD, SEM-EDS, XPS, XRF, UV-Vis spectroscopy, fluorescence spectroscopy, FT-IR, N₂ sorption isotherm and BET theory
- **Scientific computing:** Python, MATLAB, R; statistical analysis, modelling, principal component analysis; familiar with Reverse Monte-Carlo calculations, Bayesian inference, basic machine learning; Unix, Bash
- **Software:** Microsoft Office; scientific software (Athena + Artemis, XrayLarch, Thermo Advantage, Feff9, GSAS-II, Jana2006, SasView, SASfit, VESTA, Inca, Aztec, ImageJ); citation managers (Zotero, EndNote)
- **Languages:** fluent in English, native proficiency in Russian

ADVANCED COURSEWORK AND PROFESSIONAL DEVELOPMENT

<i>Artificial Intelligence in Chemistry and Materials Science</i>	Kurnakov Institute (Moscow)	Jul. 2025
<i>Synchrotron and Neutron Methods Workshop</i>	Moscow State University (Moscow)	Dec. 2024
<i>Management for Scientists and Engineers Certificate</i>	Kellogg School of Business (Evanston)	Aug. 2023
<i>Scientific Writing for the General Audience</i>	Medill School of Journalism (Evanston)	May 2023
<i>Statistical Methods for Physicists and Astronomers</i>	Northwestern University (Evanston)	Dec. 2022
<i>X-ray Diffraction Methods in Materials Science</i>	Northwestern University (Evanston)	May 2019
<i>Responsible Conduct of Research Training</i>	Northwestern University (Evanston)	Dec. 2018

HOBBIES AND INTERESTS

Aquarium and Terrarium Keeping 2019 – 2023

- Maintained a 100-litre freshwater tank housing 5 different species of fish; regularly tested water quality with qualitative colourimetric tests; applied relevant chemistry; maintained zero levels of ammonia and nitrite
- Maintained a 200-litre terrarium housing a ball python (1 meter long); ensured the animal's health by maintaining 80%+ humidity and installing two light fixtures supplying infrared and UV light

Creative Writing – Science Fiction 2021 onwards

- Took grand-prix in 3 online Russian short form fiction contests; reached finals in 4 more
- Published short form science fiction stories in paperback and in online publications

OTHER LINKS AND PROJECTS

GitHub: github.com/PexcTa

FREELANCE EXPERIENCE

Freelancer – Editor, Tutor, Reviewer

Summary: Self-employed in editor & reviewer roles; provided editorial and consulting services.

Key Achievements:

- **Jan. 2026 – Ongoing:** worked remotely at Mercor (USA) as a freelance Chemistry Expert, training an AI assistant on chemistry-themed conversations and puzzles; edited AI output for accuracy and precision
- **Oct. 2025:** Served as an invited scientific expert and reviewed a popular science article on metal–organic frameworks, providing detailed feedback through draft annotation in Word
- **Jun. 2024:** Served as freelance play tester for in-development videogame *DiscoveryWorld*; documented gameplay experience, solved puzzles, and provided constructive feedback and improvement suggestions
- **Sep. 2019 – Dec. 2022:** provided casual tutoring services within the Department of Chemistry at Northwestern University, helping 3 students with general chemistry and introductory inorganic chemistry

PUBLICATIONS

Kuzenkova, A. S.; Plakhova, T. V.; Svetogorov, R. D.; Kulikova, E. S.; Trigub, A. L.; **Kramar, B. V.**; Yapaskurt, V. O.; Toropov, A. S.; Shaulskaya, M. D.; Tsymbarenko, D. M.; Romanchuk, A. Y.; Kalmykov, S. N. Np(V) Double Carbonate with Magnesium: Synthesis, Solubility and Transformation. *Dalton Trans.* 2025.

<https://doi.org/10.1039/D5DT00913H>.

Kramar, B.V., Plakhova, T.V., Kuzenkova, A.S., Trigub, A.L., Svetogorov, R.D., Shiryayev, A.A., Nevolin, I.M., Yapryntsev, A.D., Baranchikov, A.E., Peters, G.S., Yapaskurt, V.O., Egorov, A.V., Kostyukov, I.A., Shaulskaya, M.D., Tsymbarenko, D.M., Romanchuk, A.Y., Kalmykov, S.N., 2025. Formation of a new hydrated sodium–thorium phosphate from thorium dioxide and its subsequent phase evolution. *Dalton Trans.* **2025**, 54, 7360–7375.

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Kramar, B. V.; Bondarenko, A. S.; Koehne, S. M.; Diroll, B. T.; Wang, X.; Yang, H.; Schanze, K. S.; Chen, L. X.; Tempelaar, R.; Hupp, J. T. Unexpected Photodriven Linker-to-Node Hole Transfer in a Zirconium-Based Metal–Organic Framework. *J. Phys. Chem. Lett.* **2024**, 15 (46), 11496–11503.

<https://doi.org/10.1021/acs.jpcllett.4c02848>.

Sarma, P. V.; **Kramar, B. V.**; Chen, L.; Sasmal, S.; Weingartz, N. P.; Huang, J.; Mitchell, J. B.; Kwak, M.; Chen, L. X.; Boettcher, S. W. Local Electric Field Effects on Water Dissociation in Bipolar Membranes Studied Using Core–Shell Catalysts. *Chem. Mater.* **2024**, 36 (24), 11863–11872. <https://doi.org/10.1021/acs.chemmater.4c02190>.

Elvington, M. C.; Ganesan, P.; Ward, P. A.; Liu, J.; Atilgan, A.; **Kramar, B. V.**; More, K.; Cullen, D.; Hupp, J. T.; Greenway, S.; Taylor Adams, W.; Colón-Mercado, H. R. Highly Active Oxygen Reduction Electrocatalysts Derived from an Iron-Porphyrin Framework. *PRX Energy* **2023**, 2 (4), 043008.

<https://doi.org/10.1103/PRXEnergy.2.043008>.

Liu, J.; Prelesnik, J. L.; Patel, R.; **Kramar, B. V.**; Wang, R.; Malliakas, C. D.; Chen, L. X.; Siepmann, J. I.; Hupp, J. T. A Nanocavitation Approach to Understanding Water Capture, Water Release, and Framework Physical Stability in Hierarchically Porous MOFs. *J. Am. Chem. Soc.* **2023**, 145 (51), 27975–27983.

<https://doi.org/10.1021/jacs.3c07624>.

Zheng, X.; Drummer, M. C.; He, H.; Rayder, T. M.; Niklas, J.; Weingartz, N. P.; Bolotin, I. L.; Singh, V.; **Kramar, B. V.**; Chen, L. X.; Hupp, J. T.; Poluektov, O. G.; Farha, O. K.; Zapol, P.; Glusac, K. D. Photoreactive Carbon Dioxide Capture by a Zirconium–Nanographene Metal–Organic Framework. *J. Phys. Chem. Lett.* **2023**, 14 (18), 4334–4341.

<https://doi.org/10.1021/acs.jpcllett.3c00049>.

Sheridan, T. R.; Gaidimas, M. A.; **Kramar, B. V.**; Goswami, S.; Chen, L. X.; Farha, O. K.; Hupp, J. T. Noncovalent Surface Modification of Metal–Organic Frameworks: Unscrambling Adsorption Properties via Isothermal Titration Calorimetry. *Langmuir* **2022**, 38 (37), 11199–11209. <https://doi.org/10.1021/acs.langmuir.2c01223>.

Kramar, B. V.; Flanders, N. C.; Helweh, W.; Dichtel, W. R.; Hupp, J. T.; Chen, L. X. Light Harvesting Antenna Properties of Framework Solids. *Acc. Mater. Res.* **2022**, 3 (11), 1149–1159.

<https://doi.org/10.1021/accounts.2c00137>.

Kramar, B. V.; Phelan, B. T.; Sprague-Klein, E. A.; Diroll, B. T.; Lee, S.; Otake, K.; Palmer, R.; Mara, M. W.; Farha, O. K.; Hupp, J. T.; Chen, L. X. Single-Atom Metal Oxide Sites as Traps for Charge Separation in the Zirconium-Based Metal–Organic Framework NDC–NU-1000. *Energy Fuels* **2021**, 35 (23), 19081–19095.

<https://doi.org/10.1021/acs.energyfuels.1c02623>.

CONFERENCE TALKS

Kramar, B. V.; Plakhova T. V.; Kuzenkova A. S.; Trigub A. L.; Romanchuk A. Yu.; Kalmykov S. N.; EXAFS analysis sheds light on the relationship between precipitation conditions and structural properties of double actinide carbonates and phosphates. *XXII Mendeleev Congress*, Sirius Federal Territory, Sochi, Russia, October 2024 (poster).

Kramar, B. V.; Plakhova T. V.; Kuzenkova A. S.; Trigub A. L.; Romanchuk A. Yu.; Kalmykov S. N.; Utilizing EXAFS spectroscopy to characterize structural properties of solid actinide-containing phases. *Synchrotron Radiation and Free Electron Lasers – 2024*, Budker Institute of Nuclear Physics, Novosibirsk, Russia, June 2024 (poster).

Kramar, B. V.; Phelan, B. T.; Sprague-Klein, E. A.; Diroll, B. T.; Lee, S.; Otake, K.-i.; Palmer, R.; Mara, M. W.; Farha, O. K.; Hupp, J. T.; et al. Single-Atom Metal Oxide Sites as Traps for Charge Separation in the Zirconium-Based Metal–Organic Framework NDC–NU-1000. *American Chemical Society National Meeting*, Chicago IL, August 2022 (poster).

REFEREES

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